Just use Colab, give up on VSC

PyLab5_MLmodels.ipynb is for Problem 2 of HW
PyLab3_1_causal.ipynb is for Problem 1 of HW

Use lab 3 for Q1
Use AIC, lags 1-5, EWMA

In Granger test first variable is endogenous variable, second variable is causal variable

Use 2 labs of lab 5 for Q2
deep and narrow NNs may generalize better, experiment with wide, shallow narrow
Don’t data leak or overfit
Most logical/honest is returns but hardest to model
classification problem easiest
focus on process, preprocessing, hyperparameter tuning, etc. not on results
Price is non stationary (but good results) --> not mean anything. Not use this. Use Returns (but hardest to model). Or a classification model. Results are not that important, it's about the discussion. It's important to do a good feature selection, pre-processing, hyperparameter tuning, use grid search.

Try different non-linear functions for NNs
You should never do random cross validation, split sample into train/test, more sophisticated is rolling/expanding window

Always plot everything to check, document and justify all decisions for tuning INSIDE the code

Small alphas tend to overfit- > use validation set (need to split into train, validation and test sets if want to get fancy)

For GNs, experiment with different Kernels

For Gaussian N need the parameters to converge?? Check convergence warnings, fix bounds/make them wider maybe
Better to use returns
Can choose EITHER alpha or wide kernel, alpha is better but both work, just don’t use both

Good results in training + bad in testing = overfitting
Overfitting in GNs happens due to small alpha, we assume no noise! Very bad for financial data, either switch to wide kernel or try bigger alpha, can do grid search for optimal alpha

Focus on tuning, finding best parameters, experiment with different parameters.

LAB 2 GOOD:

Building network from causality
Time series split
For LSTM/GP: `GridSearchCV` does not wrap a Keras LSTM cleanly anymore (write a manual loop over `(timesteps, units, layers)` using the same `TimeSeriesSplit` folds).

For GP, grid-search the kernel choice and `alpha`; the kernel hyperparameters are optimized inside `fit()` via marginal likelihood.

Bootstrapping:
Kind of optional for HW, don’t stress, can be used for Feature Engineering
Basic bootstrap can not be applied for financial time series
Block bootstrap takes blocks, also suffers from same problem, both produce non-stationary samples

Stationary bootstrap produces stationary samples